CHEST

The Impact of the 2023 American Cancer Society Screening Recommendations on Racial, Ethnic, and Sex Disparities in Lung Cancer Screening Eligibility

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BACKGROUND: Because lung cancer risk remains elevated beyond 15 years after smoking cessation, the American Cancer Society (ACS) recommended in 2023 to remove the quit duration criterion from the US Preventive Services Task Force (USPSTF) lung cancer screening guidelines.

RESEARCH QUESTION: How do the ACS recommendations impact racial, ethnic, and sex disparities in lung screening eligibility?

STUDY DESIGN AND METHODS: We identified adults who currently and formerly smoked from the Black Women's Health Study, Multiethnic Cohort Study, and Southern Community Cohort Study recruited between 1993 and 2009 and followed up through 2019 for lung cancer incidence. Data across cohorts were harmonized. We evaluated the performance of the USPSTF guidelines and ACS recommendations across race, ethnicity, and sex.

RESULTS: Among 175,259 adults with a smoking history included (42% Black/African American [AA], 25% White American, 15% Japanese American, 12% Hispanic or Latino, 5% Native Hawaiian, and 1% multiracial or other race), 27% (n = 46,702) were eligible for screening according to USPSTF guidelines and 33% (n = 58,373) were eligible according to ACS recommendations. Among those eligible with ACS recommendations but not with USPSTF guidelines (n = 11,671 individuals who formerly smoked), 382 participants (3%) developed lung cancer. ACS recommendations increased eligibility sensitivity, but decreased eligibility specificity among all participants who formerly smoked (USPSTF: sensitivity, 0.34 [95% CI, 0.32-0.35]; specificity, 0.87 [95% CI, 0.87-0.87]; ACS: sensitivity, 0.44 [95% CI, 0.42-0.45]; specificity, 0.77 [95% CI, 0.77-0.77]). These changes differed in magnitude across racial and ethnic groups, resulting in larger eligibility disparities. White American individuals who formerly smoked and developed lung cancer were 3 to 10 percentage points more likely to be eligible for screening with ACS vs USPSTF when compared with their non-White counterparts. The increased disparity was most evident between Black or AA individuals and White American individuals (USPSTF disparity, 7 percentage points; ACS disparity, 12 percentage points) and between Hispanic or Latino and White American individuals (USPSTF disparity, 12 percentage points; ACS disparity, 22 percentage points).

INTERPRETATION: ACS recommendations increased screening eligibility regardless of cancer status, potentially leading to unnecessary screening of people without lung cancer. Despite increased eligibility sensitivity, changes were not uniform across groups, resulting in larger disparities in race and ethnicity and in sex. CHEST 2026; 169(6):1750-1763

KEY WORDS: ACS; lung cancer; lung cancer screening guidelines; racial disparities; screening eligibility; USPSTF

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Take-Home Points

Research Question: Are existing racial, ethnic, and sex disparities in lung cancer screening eligibility impacted by the 2023 American Cancer Society (ACS) recommendation to remove the quit duration eligibility criterion?

Results: Compared with the US Preventive Services Task Force 2021 guidelines, the ACS recommendations disproportionately increased eligibility among White American individuals who formerly smoked in comparison with their counterparts of Japanese American, Native Hawaiian, Black or African American, or Hispanic or Latino descent.

Interpretation: The ACS recommendations led to larger racial, ethnic, and sex disparities in eligibility regardless of lung cancer status and potentially increased unnecessary screening of people without lung cancer.

Lung cancer is the leading cause of cancer death in the United States.[1] Since 2013, the US Preventive Services Task Force (USPSTF) has endorsed annual screening with low-dose CT (LDCT) imaging, which has been shown to reduce lung cancer mortality among high-risk individuals.[2] These screening guidelines are based on 3 criteria: age, smoking history, and quit duration. The initial guidelines recommended annual LDCT imaging for individuals between 55 and 80 years of age with a minimum 30-pack-year smoking history who either currently smoke or who quit within the last 15 years.[3] Subsequent analyses by our team and others demonstrated racial disparities in these screening eligibility guidelines.[4-6] As a result, USPSTF revised the criteria in 2021 by lowering the minimum screening age to 50 years and lowering the smoking history threshold to a minimum of 20 pack-years.[7] Although these changes were intended to include more female and Black or African American (AA) people for screening, racial disparities in screening eligibility persist.[8,9] In November 2023, the American Cancer Society (ACS) recommended further revision to the USPSTF 2021 guidelines to remove the quit duration criterion.[10] This change was based on evidence that individuals who formerly smoked maintain an elevated risk of lung cancer developing beyond 15 years since quitting and was intended to expand eligibility to include this high-risk group.[11-13] Herein, we compare the performance of the ACS recommendations with that of the current USPSTF 2021 guidelines and evaluate their impact on race, ethnicity, and sex disparities in screening eligibility.

Study Design and Methods

We identified study participants from 3 large, racially diverse prospective cohort studies: the Black Women's Health Study (BWHS), the Multiethnic Cohort Study (MEC), and the Southern Community Cohort Study (SCCS). The BWHS recruited 59,000 Black American female individuals 21 to 69 years of age via mailed questionnaires to subscribers of *Essence* magazine in 1995. The MEC enrolled about 215,000 adults 45 to 75 years of age recruited primarily through driver's license files from California and Hawaii between 1993 and 1996. The SCCS enrolled about 85,000 English-speaking adults 40 to 79 years of age recruited predominantly from community health centers in 12 southeastern states between 2002 and 2009. Participants from all 3 cohorts were followed up through 2019 for health outcomes. Complete details on recruitment, inclusion, and exclusion criteria for each study are available elsewhere.[14-17] All studies collected comprehensive questionnaires on demographic characteristics, smoking behaviors, and relevant exposures throughout study follow-up. A total of 13,831 participants (1,673 from

ABBREVIATIONS: AA = African American; ACS = American Cancer Society; BWHS = Black Women's Health Study; LDCT = low-dose CT; LR = likelihood ratio; MEC = Multiethnic Cohort; SCCS = Southern Community Cohort Study; USPSTF = US Preventive Services Task Force

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the SCCS, 11,983 from the MEC, and 175 from the BWHS) with incomplete smoking or race and ethnicity data, or both, were excluded from analyses. Data were harmonized and pooled into a cohort of 175,259 participants with either active tobacco use or a prior smoking history. e-Figure 1 shows the study flowchart. The study was approved by Vanderbilt University Medical Center (Identifier: 210269).

Participants with incident lung cancer were identified from cancer registries or medical records confirmed by pathology. For each participant, screening eligibility was determined using the USPSTF 2021 guidelines[7] or the ACS 2023 recommendations[10] based on smoking behaviors at the time of diagnosis for participants with lung cancer or the most recent follow-up for participants without lung cancer (e-Table 1). Pack-years of smoking were calculated as the number of cigarettes smoked per day divided by 20 (number of cigarettes per pack) and multiplied by the duration of smoking. For participants who reported current smoking at baseline but who were missing follow-up smoking information, we estimated pack-years assuming that the last recorded smoking status and number of cigarettes smoked per day remained the same. Quit duration was calculated as the duration from smoking cessation until lung cancer diagnosis or the most recent follow-up. e-Table 2 shows examples of the questions used to derive these variables at baseline and follow-up for each study. Participants were categorized into subgroups based on sex (male or female) and self-reported race and ethnicity (non-Hispanic Black or AA, non-Hispanic White American, Hispanic or Latino, Japanese American, Native Hawaiian, and multiracial or other race or ethnicity, which included participants who identified as American Indian, Alaskan Native, multiple races, or other race).

Statistical Analysis

Eligibility determinations from USPSTF guidelines and ACS recommendations were compared across race, ethnicity, and sex. Eligibility sensitivity was defined as the proportion of participants eligible for screening among those who developed lung cancer, that is: P(eligible for screening | lung cancer diagnosed). Eligibility specificity was defined as the proportion of participants ineligible for screening among those who did not develop lung cancer, that is: P(ineligible for screening | no lung cancer diagnosed). Positive predictive value was defined as the proportion of eligible participants who developed lung cancer, that is: P(lung cancer | eligible for screening). The positive likelihood ratio (LR) was defined as the ratio of eligible participants with lung cancer to eligible participants without lung cancer and can be interpreted as a measure of how well each guideline captures individuals at risk of developing lung cancer. The cancer detection rate was defined as the proportion of participants who developed lung cancer who were eligible for screening among all participants eligible for screening, that is:

$$\left(\text{Sensitivity} \times \frac{\text{Total No. of Participants With Lung Cancer}}{\text{Total No. of Participants Eligible for Screening}}\right)$$

and measures how well each guideline identifies people with lung cancer among those eligible for screening. The number needed to detect is the number of participants required to be eligible to identify 1 person correctly who would develop lung cancer for screening and was calculated as the reciprocal of the cancer detection rate: 1 / cancer detection rate. To assess the impact of ACS recommendations on screening eligibility, we calculated the change in sensitivity and in specificity (ACS – USPSTF). This was evaluated overall and within self-reported race (eg, Black or AA – White American), ethnicity (Hispanic or Latino – White American), and sex (female – male) subgroups. Comparison between selected subgroups, that is, the change in eligibility sensitivity among male participants vs the change in eligibility sensitivity among female participants, is the disparity differential resulting from implementing the ACS recommendations. Self-reported White race and male sex were used as the reference groups in this analysis to highlight the impact of recommendations on underrepresented groups. Differences in sensitivity and specificity (ie, accuracy changes between guidelines) were assessed using difference-in-proportion tests with estimates and 95% CIs reported. The 95% CI for the disparity change between guidelines was estimated using bootstrapping. We resampled 200 data sets with replacement and with sample size equal to the total study population. All analyses were completed using R version 4.3.1 software (R Foundation for Statistical Computing).

Results

Among 175,259 adults with a smoking history, 73,603 participants (42%) self-identified as Black or AA, 44,253 participants (25%) identified as White American, 26,362 participants (15%) identified as Japanese American, 21,488 participants (12%) identified as Hispanic or Latino, 7,961 participants (5%) identified as Native Hawaiian, and 1,546 participants (1%) identified as being multiracial or from another race. A total of 7,655 participants (4.4%) developed lung cancer over the 26 years of follow-up. This percentage differed by subgroup, ranging from 3% among multiracial or other race participants to 7% among Native Hawaiian participants. About two-thirds of the cohort reported formerly smoking, with median quit durations ranging from 18 years among Native Hawaiian and Hispanic or Latino participants to 21 years among Black or AA, White American, and Japanese American participants. The median pack-years of smoking was 16 pack-years (Table 1).

When the USPSTF guidelines were applied, 27% of participants who had ever smoked (n = 46,702) were eligible for screening. This figure ranged from 14% among Hispanic and Latino participants to 46% among participants who identified as multiracial or of other race (Table 2). After applying the ACS recommendations, 33% of participants who had ever smoked (n = 58,373) were eligible for screening. As expected, the additional 6% of participants (n = 11,671) who were eligible with ACS recommendations but not with USPSTF guidelines were all individuals who formerly smoked. Of those, 382 participants (3%) developed lung cancer, among whom 247 participants (65%) were male (25% White American, 20% Japanese American, 11% Black or AA, 5% Native Hawaiian, and 3% Hispanic) and 135 participants (35%) were female (15% White American, 2% Japanese American, 14% Black or AA, 2% Native Hawaiian, and 1% Hispanic).

Compared with the USPSTF guidelines, the ACS recommendations increased eligibility sensitivity, identifying more participants for screening who later developed lung cancer. However, the ACS recommendations also decreased eligibility specificity compared with the USPSTF guidelines, identifying more participants for screening who did not develop lung cancer. Table 3 shows these changes in sensitivity and specificity by race, ethnicity, and sex subgroups among participants who formerly smoked. Importantly, the magnitude of these changes varied by subgroup. White American participants who formerly smoked showed a larger absolute increase in sensitivity with ACS vs USPSTF guidelines (accuracy change, 0.14 [95% CI, 0.10-0.18]) compared with Native Hawaiian, Japanese American, and Black or AA participants (accuracy change: 0.11 [95% CI, 0.03-0.20], 0.10 [95% CI, 0.05-0.14], and 0.09 [95% CI, 0.05-0.13], respectively). Similarly, White American participants who formerly smoked showed a larger absolute decrease in specificity with ACS vs USPSTF guidelines (accuracy change, −0.15 [95% CI, −0.16 to −0.14]) compared with Native Hawaiian, Japanese American, Black or AA, and Hispanic or Latino participants (accuracy change: −0.09 [95% CI, −0.10 to −0.07], −0.09 [95% CI, −0.10 to −0.08], −0.10 [95% CI, −0.11 to −0.10], and −0.04 [95% CI, −0.05 to −0.03], respectively) (Table 3). Male individuals who formerly smoked also showed larger absolute changes in accuracy (accuracy change: sensitivity, 0.12 [95% CI, 0.09-0.15]; specificity, −0.12 [95% CI, −0.13 to −0.12]) compared with female individuals who formerly smoked (accuracy change: sensitivity, 0.08 [95% CI, 0.05-0.12]; specificity, −0.09 [95% CI, −0.09 to −0.08]).

Although the magnitude of changes in sensitivity and specificity varied by subgroup, racial and ethnic disparities in screening eligibility consistently were larger for all subgroups (Fig 1, Table 3). The disparity increase was most evident between Hispanic or Latino and White American individuals. For every 100 participants who developed cancer assessed with USPSTF guidelines, 12 more White American participants were eligible for screening compared with Hispanic or Latino participants (within-race ethnicity disparity by USPSTF guidelines: sensitivity, −0.12 [95% CI, −0.17 to −0.07]). However, for every 100 participants who developed cancer assessed with the ACS recommendations, 22 more White American participants were eligible for screening compared with Hispanic or Latino participants (within-race ethnicity disparity by ACS recommendations: sensitivity, −0.22 [95% CI, −0.27 to −0.16]). That is, applying the ACS recommendations led to a 10-percentage point increase in the eligibility disparity between White American and Hispanic or Latino participants. These worsening disparities also were evident in some subgroups defined by race, ethnicity, and sex. They ranged from a 4-percentage point increase in disparity between Native Hawaiian and White American female individuals to a 12-percentage point increase in disparity between

TABLE 1] Demographic Characteristics and Smoking Histories of All Study Participants by Race and Ethnicity

Characteristic	Black or African American (n = 73,605)	White American (n = 44,253)	Japanese American (n = 26,362)	Hispanic or Latino (n = 21,488)	Native Hawaiian (n = 7,961)	Multiracial or Other (n = 1,546)	Total (N = 175,259)
Incident lung cancer	2,809 (3.8)	2,177 (4.9)	1,333 (5.1)	729 (3.4)	557 (7.0)	50 (3.1)	7,655 (4.4)
Sex							
Female	48,158 (65.4)	22,769 (51.5)	8,471 (32.1)	7,648 (35.6)	3,990 (50.1)	924 (58.1)	91,960 (52.5)
Male	25,447 (34.6)	21,484 (48.5)	17,891 (67.9)	13,840 (64.4)	3,971 (49.9)	666 (41.9)	83,299 (47.5)
Current age, y[a]	65.0 (23.0-100)	73.2 (40.0-100)	80.2 (45.6-100)	79.2 (36.0-100)	74.5 (45.8-99.6)	61.0 (41.0-92.0)	72.5 (23.0-100)
Age at diagnosis, y[b]	68.0 (24.0-95.2)	71.9 (40.0-96.5)	76.3 (49.0-96.3)	74.8 (49.3-94.6)	71.4 (46.2-98.3)	61.0 (48.0-83.0)	71.9 (24.0-98.3)
Highest education level							
Less than high school	13,566 (18.4)	5,201 (11.8)	863 (3.3)	6,643 (30.9)	434 (5.5)	421 (26.5)	27,128 (15.5)
High school	22,182 (30.1)	12,401 (28.0)	9,509 (36.1)	7,558 (35.2)	4,107 (51.6)	432 (27.2)	56,189 (32.1)
More than high school	37,669 (51.2)	26,533 (60.0)	15,887 (60.3)	7,057 (32.8)	3,384 (42.5)	737 (46.4)	91,267 (52.1)
Missing	188 (0.3)	118 (0.3)	103 (0.4)	230 (1.1)	36 (0.5)	0 (0)	675 (0.4)
Family history of lung cancer	6,556 (8.9)	5,709 (12.9)	2,904 (11.0)	1,455 (6.8)	1,033 (13.0)	203 (12.8)	17,860 (10.2)
BMI, kg/m^2	28.0 (11.3-84.5)	26.2 (9.76-74.7)	24.2 (9.76-57.1)	27.0 (11.0-81.8)	27.5 (9.01-66.6)	28.3 (15.5-67.3)	26.6 (9.01-84.5)
Current smoking status[a]							
Current smoking	34,590 (47.0)	14,840 (33.5)	4,216 (16.0)	4,682 (21.8)	2,142 (26.9)	998 (62.8)	61,468 (35.1)
Former smoking	39,015 (53.0)	29,413 (66.5)	22,146 (84.0)	16,806 (78.2)	5,819 (73.1)	592 (37.2)	113,791 (64.9)
No. of cigarettes/d	8.00 (0-200)	15.5 (0.500-200)	15.5 (1.00-50.0)	8.00 (0-200)	15.5 (1.00-60.0)	18.0 (0.500-140)	10.0 (0-200)
Current smoking duration, y[a]	29.0 (0-78.0)	25.5 (0-83.0)	25.5 (5.00-51.0)	20.0 (0-63.0)	25.5 (5.00-51.0)	38.0 (0-76.0)	25.5 (0-83.0)
Current pack-years of smoking, y[a]	14.2 (0-312)	24.0 (0.0125-470)	19.8 (1.25-105)	8.88 (0-460)	19.8 (1.25-84.1)	25.0 (0.0125-210)	16.4 (0-470)
Current quit duration, y[a,c]	21.0 (0-73.0)	21.0 (0-73.0)	21.0 (0.500-31.0)	18.0 (0-63.0)	18.0 (0.500-31.0)	25.0 (0-66.0)	19.0 (0-73.0)

Data are presented as No. (%) or median (range).
[a]Current indicates at the time of diagnosis for those with lung cancer or at time of most recent follow-up for those without lung cancer.
[b]Among participants with lung cancer.
[c]Among individuals who formerly smoked.

TABLE 2] Measures of Eligibility Accuracy of USPSTF 2021 Guidelines and ACS 2023 Recommendations for All Study Participants

Measure	USPSTF 2021	ACS 2023
Total eligible	46,702 (26.6)	58,373 (33.3)
Black or African American	19,675 (26.7)	23,627 (32.1)
White American	15,733 (35.6)	20,145 (45.5)
Japanese American	5,501 (20.9)	7,462 (28.3)
Hispanic or Latino	2,973 (13.8)	3,662 (17.0)
Native Hawaiian	2,095 (26.3)	2,600 (32.7)
Multiracial or other	725 (45.6)	877 (55.2)
Eligibility sensitivity[a]	0.47 (0.46-0.49)	0.52 (0.51-0.53)
Eligibility specificity[b]	0.74 (0.74-0.75)	0.68 (0.67-0.68)
Positive predictive value[c]	0.08 (0.07-0.08)	0.07 (0.07-0.07)
Positive likelihood ratio[d]	1.84 (1.80-1.89)	1.61 (1.58-1.65)
Cancer detection rate per 1,000[e]	78 (76-80)	69 (67-70)
No. needed to detect[f]	13 (13-14)	15 (15-15)

Data are presented as No. (%) or value (95% CI). ACS = American Cancer Society; USPSTF = United State Preventive Task Force.
[a]Defined as the proportion of participants eligible for screening among those who developed lung cancer: P(eligible for screening | lung cancer).
[b]Defined as the proportion of participants ineligible for screening among those who did not develop lung cancer: P(ineligible for screening | no lung cancer).
[c]Defined as the proportion of eligible participants who developed lung cancer: P(lung cancer | eligible for screening).
[d]Defined as the ratio of eligible participants with lung cancer to eligible participants without lung cancer.
[e]Measured the proportion of participants with lung cancer who were eligible for screening among all participants eligible for screening scaled per 1,000 people: sensitivity × (total No. of participatns with lung cancer / total No. of participants eligible for screening) × 1,000. 95% CIs estimated using 2.5th and 97.5th quantiles of 200 bootstrap samples.
[f]Measured the number of participants that needed to be eligible for screening to identify correctly 1 person with lung cancer and was calculated as the reciprocal of the detection rate: (1 / cancer detection rate). 95% CIs estimated using 2.5th and 97.5th quantiles of 200 bootstrap samples.

Hispanic or Latino and White American male individuals.

The screening efficiency seemed to decline with the ACS recommendations. Removing the quit duration criterion effectively changed the true-positive to false-positive trade-off without improving eligibility accuracy. This changing trade-off was captured by the positive LR curve which, in this context, displayed the likelihood of lung cancer developing among those eligible for screening. A visual inspection of the positive LR curves (Fig 2) shows that in all cases, a person eligible for screening with the ACS recommendations (which had a lower positive LR) was less likely to develop lung cancer than a person eligible with the current USPSTF guidelines. This is supported by the lower cancer detection rate of the ACS recommendations compared with the USPSTF guidelines. For every 1,000 participants eligible for screening, the USPSTF guidelines identified 9 more participants (95% CI, 9-10 more participants) who developed lung cancer for screening compared with the ACS recommendations. This is also supported by the higher number needed to detect of the ACS recommendations compared with the USPSTF guidelines, which suggests that, if implemented, the ACS recommendations would require more people to be eligible for screening to identify 1 case of lung cancer (Table 2).

Discussion

Although the ACS recommendations were not aimed at addressing racial and ethnic disparities, applying them did not seem to improve equity with regard to screening eligibility. Varying changes in eligibility led to wider disparities especially between Black or AA and White male individuals and between Hispanic or Latino and White male individuals. Our findings highlight the importance of validating the performance of new guidelines in more diverse settings to understand their impact further when implemented. We found that removing the quit duration criterion increased sensitivity and decreased specificity differentially across subgroups, resulting in larger racial and ethnic disparities in eligibility despite increasing the absolute number of lung cancer cases identified.

The evidence in support of the ACS recommendations was drawn from predominantly White American populations (eg, the Framingham Heart Study, the

TABLE 3] Sensitivity, Specificity, and Differentials of USPSTF 2021 Guidelines and ACS 2023 Recommendations by Race, Ethnicity, and Sex Only Among Participants Who Reported Formerly Smoking

Demographic	No.	USPSTF 2021		ACS 2023		Accuracy Change[c]		Disparity Change[d]	
		Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity Differential[a]	Specificity Differential[b]	Sensitivity Differential[a]	Specificity Differential[b]
Former smoking	113,791	0.34 (0.32 to 0.35)	0.87 (0.87 to 0.87)	0.44 (0.43 to 0.45)	0.77 (0.77 to 0.77)	0.10 (0.08 to 0.12)	−0.10 (−0.10 to −0.11)		
By race and ethnicity									
White	29,413	0.39 (0.36 to 0.42)	0.83 (0.83 to 0.84)	0.53 (0.50 to 0.56)	0.68 (0.68 to 0.69)	0.14 (0.10 to 0.18)	−0.15 (−0.16 to −0.14)		
Black or African American	39,015	0.32 (0.30 to 0.35)	0.90 (0.90 to 0.90)	0.41 (0.38 to 0.44)	0.80 (0.79 to 0.80)	0.09 (0.05 to 0.13)	−0.10 (−0.11 to −0.10)		
Japanese American	22,146	0.31 (0.28 to 0.34)	0.86 (0.85 to 0.86)	0.41 (0.37 to 0.44)	0.77 (0.76 to 0.77)	0.10 (0.05 to 0.14)	−0.09 (−0.10 to −0.08)		
Hispanic or Latino	16,809	0.27 (0.23 to 0.32)	0.91 (0.91 to 0.92)	0.31 (0.27 to 0.36)	0.87 (0.87 to 0.88)	0.04 (−0.02 to 0.11)	−0.04 (−0.05 to −0.03)		
Native Hawaiian	5,819	0.32 (0.26 to 0.38)	0.82 (0.81 to 0.83)	0.43 (0.36 to 0.49)	0.73 (0.72 to 0.74)	0.11 (0.03 to 0.20)	−0.09 (−0.10 to −0.07)		
Multiracial or other[e]	568	0.36 (0.13 to 0.65)	0.84 (0.81 to 0.87)	0.64 (0.35 to 0.87)	0.59 (0.55 to 0.63)	0.29 (−0.07 to 0.64)	−0.26 (−0.31 to −0.21)		
Within-race and ethnicity disparity[f]									
Black or African American vs White		−0.07 (−0.11 to −0.03)	0.07 (0.06 to 0.07)	−0.12 (−0.16 to −0.08)	0.11 (0.11 to 0.12)			−0.06 (−0.08 to −0.03)	0.05 (0.04 to 0.05)
Japanese American vs White		−0.08 (−0.12 to −0.04)	0.02 (0.02 to 0.03)	−0.13 (−0.17 to −0.08)	0.09 (0.08 to 0.09)			−0.05 (−0.07 to −0.02)	0.06 (0.06 to 0.07)
Hispanic or Latino vs White		−0.12 (−0.17 to −0.07)	0.08 (0.08 to 0.09)	−0.22 (−0.27 to −0.16)	0.19 (0.18 to 0.20)			−0.10 (−0.12 to −0.07)	0.11 (0.1 to 0.11)
Native Hawaiian vs White		−0.08 (−0.14 to −0.01)	−0.01 (−0.02 to 0.00)	−0.10 (−0.17 to −0.03)	0.05 (0.04 to 0.06)			−0.03 (−0.07 to 0.02)	0.06 (0.06 to 0.07)
Multiracial or other[e] vs White		−0.03 (−0.29 to 0.22)	0.01 (−0.02 to 0.04)	0.11 (−0.14 to 0.36)	−0.09 (−0.14 to −0.05)			0.14 (−0.06 to 0.40)	−0.11 (−0.14 to −0.07)
By sex									
Female	58,980	0.31 (0.29 to 0.34)	0.90 (0.89 to 0.90)	0.40 (0.37 to 0.42)	0.81 (0.81 to 0.81)	0.08 (0.05-0.12)	−0.09 (−0.09 to −0.08)		
Male	54,811	0.35 (0.33 to 0.37)	0.84 (0.84 to 0.85)	0.47 (0.45 to 0.49)	0.72 (0.72 to 0.73)	0.12 (0.09-0.15)	−0.12 (−0.13 to −0.12)		
Within-sex disparity[f]									
Female vs male		−0.04 (−0.07 to −0.01)	0.05 (0.05 to 0.06)	−0.07 (−0.10 to −0.04)	0.09 (0.08 to 0.09)			−0.03 (−0.05 to −0.02)	0.03 (0.03 to 0.04)

(Continued)

TABLE 3] *(Continued)*

Demographic	No.	USPSTF 2021		ACS 2023		Accuracy Change[c]		Disparity Change[d]	
		Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity Differential[a]	Specificity Differential[b]	Sensitivity Differential[a]	Specificity Differential[b]
By race or ethnicity and female sex									
White female	14,439	0.37 (0.33 to 0.42)	0.85 (0.84 to 0.85)	0.49 (0.44 to 0.53)	0.72 (0.72 to 0.73)	0.11 (0.05 to 0.17)	−0.12 (−0.13 to −0.11)		
Black or African American female	28,711	0.29 (0.26 to 0.33)	0.92 (0.91 to 0.92)	0.37 (0.33 to 0.41)	0.82 (0.82 to 0.82)	0.08 (0.03 to 0.13)	−0.10 (−0.10 to −0.09)		
Japanese American female	6,831	0.28 (0.22 to 0.34)	0.90 (0.90 to 0.91)	0.31 (0.25 to 0.38)	0.88 (0.87 to 0.88)	0.03 (−0.05 to 0.12)	−0.03 (−0.04 to −0.02)		
Hispanic or Latina female	5,853	0.19 (0.12 to 0.28)	0.94 (0.94 to 0.95)	0.24 (0.16 to 0.33)	0.92 (0.92 to 0.93)	0.05 (−0.06 to 0.16)	−0.02 (−0.03 to −0.01)		
Native Hawaiian female	2,782	0.36 (0.26 to 0.46)	0.84 (0.83 to 0.85)	0.43 (0.34 to 0.53)	0.79 (0.77 to 0.81)	0.08 (−0.06 to 0.21)	−0.05 (−0.07 to −0.03)		
Multiracial or other[e] female	353	0.38 (0.09 to 0.76)	0.84 (0.8 to 0.88)	0.88 (0.47 to 1.0)	0.63 (0.58 to 0.68)	0.50 (0.09 to 0.91)	−0.21 (−0.28 to −0.15)		
Within-race or ethnicity disparity (among female participants)[f]									
Black or African American female vs White female		−0.08 (−0.14 to −0.03)	0.07 (0.06 to 0.07)	−0.12 (−0.18 to −0.06)	0.10 (0.09 to 0.10)			−0.03 (−0.07 to 0.00)	0.03 (0.02 to 0.03)
Japanese American female vs White female		−0.10 (−0.17 to −0.02)	0.06 (0.05 to 0.07)	−0.18 (−0.25 to −0.10)	0.15 (0.14 to 0.16)			−0.08 (−0.12 to −0.04)	0.09 (0.09 to 0.1)
Hispanic or Latina female vs White female		−0.19 (−0.27 to −0.10)	0.10 (0.09 to 0.10)	−0.25 (−0.34 to −0.16)	0.20 (0.19 to 0.21)			−0.07 (−0.11 to −0.01)	0.10 (0.10 to 0.11)
Native Hawaiian female vs White female		−0.02 (−0.12 to 0.08)	−0.01 (−0.02 to 0.01)	−0.05 (−0.16 to 0.05)	0.07 (0.05 to 0.08)			−0.04 (−0.09 to 0.02)	0.07 (0.06 to 0.08)
Multiracial or other[e] female vs White female		0.00	−0.01 (−0.05 to 0.03)	0.39 (0.16 to 0.62)	−0.10 (−0.15 to −0.04)			0.39 (0.05 to 0.79)	−0.09 (−0.13 to −0.05)
By race or ethnicity and male sex									
White male	14,974	0.41 (0.37 to 0.45)	0.82 (0.81 to 0.82)	0.57 (0.53 to 0.61)	0.64 (0.63 to 0.65)	0.17 (0.11 to 0.22)	−0.18 (−0.19 to −0.17)		
Black or African American male	10,304	0.38 (0.33 to 0.43)	0.85 (0.84 to 0.86)	0.48 (0.43 to 0.53)	0.73 (0.72 to 0.74)	0.10 (0.03 to 0.16)	−0.12 (−0.13 to −0.11)		
Japanese American male	15,315	0.32 (0.29 to 0.36)	0.83 (0.83 to 0.84)	0.44 (0.40 to 0.48)	0.72 (0.71 to 0.73)	0.11 (0.06 to 0.17)	−0.11 (−0.12 to −0.11)		

(Continued)

TABLE 3] *(Continued)*

Demographic	No.	USPSTF 2021		ACS 2023		Accuracy Change[c]		Disparity Change[d]	
		Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity Differential[a]	Specificity Differential[b]	Sensitivity Differential[a]	Specificity Differential[b]
Hispanic or Latino male	10,953	0.30 (0.25 to 0.36)	0.90 (0.89 to 0.90)	0.34 (0.29 to 0.40)	0.85 (0.84 to 0.85)	0.04 (−0.03 to 0.12)	−0.05 (−0.06 to −0.04)		
Native Hawaiian male	3,037	0.28 (0.21 to 0.37)	0.80 (0.78 to 0.81)	0.43 (0.34 to 0.51)	0.68 (0.66 to 0.70)	0.14 (0.03 to 0.26)	−0.12 (−0.14 to −0.10)		
Multiracial or other[e] male	215	0.33 (0.04 to 0.78)	0.85 (0.8 to 0.90)	0.33 (0.04 to 0.78)	0.52 (0.45 to 0.59)	0.00 (−0.53 to 0.53)	−0.33 (−0.41 to −0.25)		
Within-race or ethnicity disparity (among male participants)[f]									
Black or African American male vs White male		−0.03 (−0.09 to 0.03)	0.03 (0.02 to 0.04)	−0.10 (−0.16 to −0.03)	0.09 (0.08 to 0.10)			−0.07 (−0.11 to −0.02)	0.06 (0.05 to 0.06)
Japanese American male vs White male		−0.08 (−0.14 to −0.03)	0.02 (0.01 to 0.02)	−0.14 (−0.19 to −0.08)	0.08 (0.07 to 0.09)			−0.05 (−0.09 to −0.01)	0.06 (0.05 to 0.07)
Hispanic or Latino male vs White male		−0.11 (−0.17 to −0.04)	0.08 (0.07 to 0.09)	−0.23 (−0.30 to −0.16)	0.21 (0.20 to 0.22)			−0.12 (−0.16 to −0.09)	0.12 (0.12 to 0.13)
Native Hawaiian male vs White male		−0.12 (−0.21 to −0.04)	−0.02 (−0.03 to 0.00)	−0.15 (−0.24 to −0.05)	0.04 (0.02 to 0.06)			−0.02 (−0.09 to 0.04)	0.06 (0.04 to 0.07)
Multiracial or other[e] male vs White male		−0.07 (−0.45 to 0.31)	0.03 (−0.01 to 0.08)	−0.24 (−0.62 to 0.14)	−0.12 (−0.19 to −0.05)			−0.17 (−0.19 to −0.14)	−0.15 (−0.22 to −0.09)

Data are presented as value (95% CI) unless otherwise indicated. ACS = American Cancer Society; USPSTF = US Preventive Task Force.
[a]Sensitivity of ACS 2023 minus sensitivity of USPSTF 2021 guidelines.
[b]Specificity of ACS 2023 minus specificity of USPSTF 2021 guidelines.
[c]Differences in sensitivity and specificity between ACS 2023 recommendations and USPSTF 2021 guidelines and calculated using difference-in-proportion tests with estimates and 95% CIs reported.
[d]Calculated as the change in accuracy between subgroups (eg, change in eligibility sensitivity among male participants vs the change in eligibility sensitivity among female participants) and measured the change in disparity as a result of applying ACS 2023 recommendations. The 95% CIs for the disparity change between guidelines were estimated using bootstrapping.
[e]Combines participants from heterogeneous backgrounds including participants who identified as American Indian, Alaskan Native, multiple races, or other race. Estimates should be interpreted with caution.
[f]Within-group disparities were calculated as the difference between referent group and other subgroup levels. Referent groups were White, male, White female, or White male.

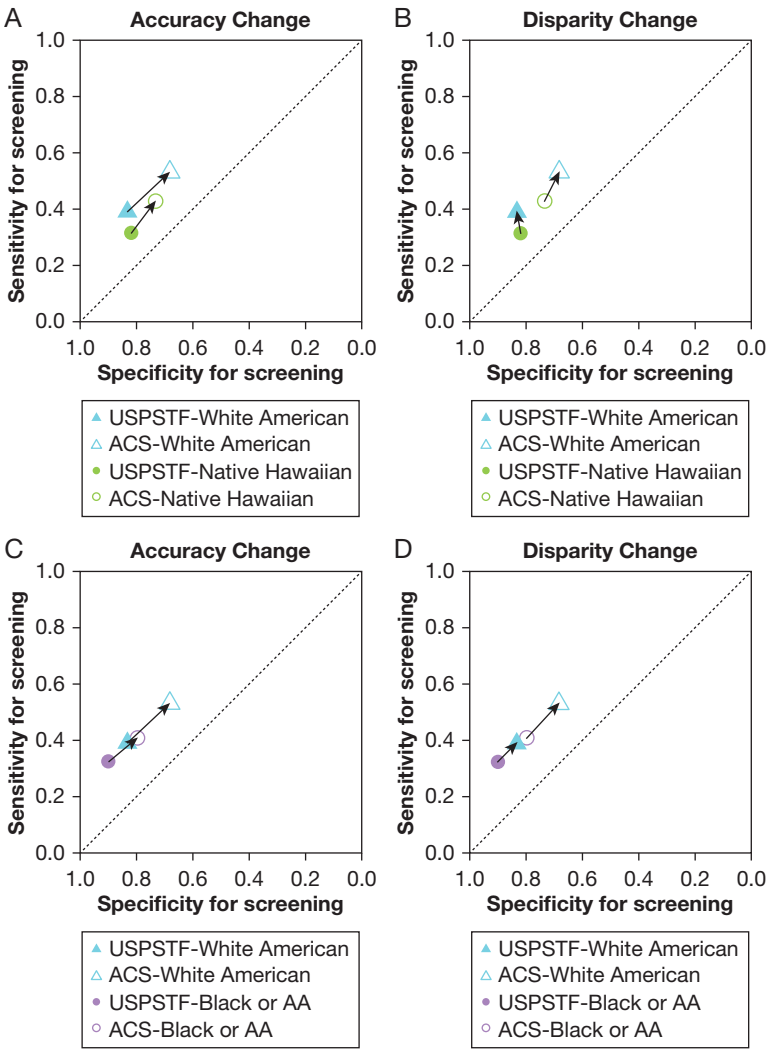


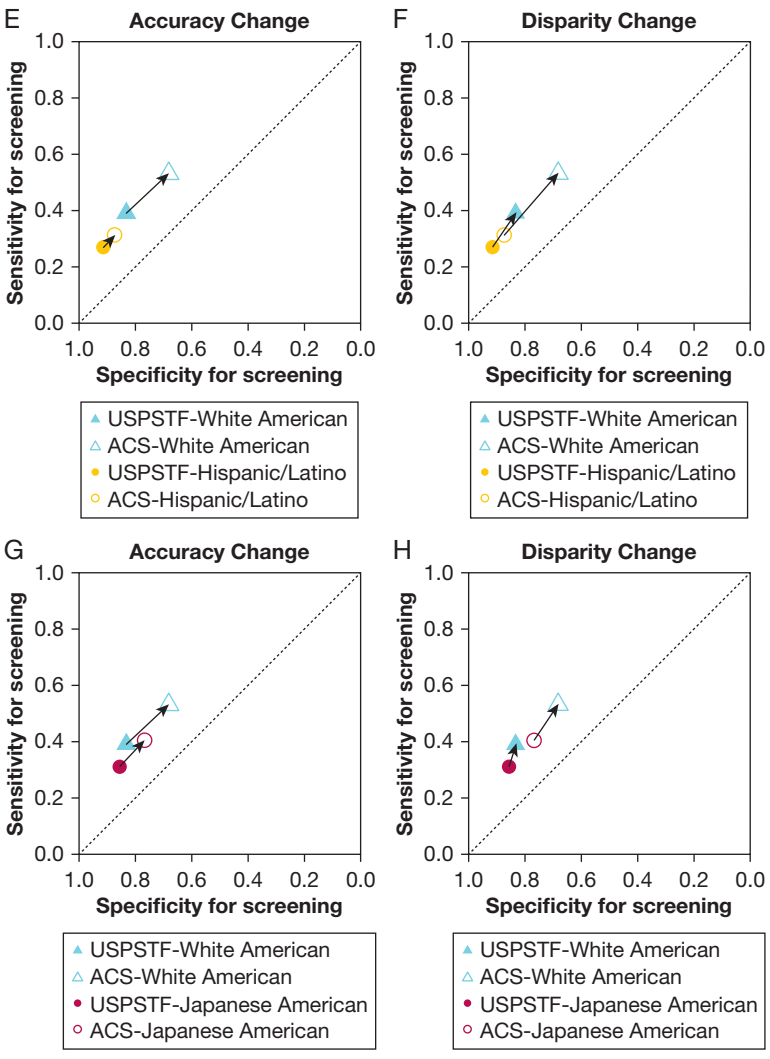
Figure 1 – .*Graphs showing accuracy and disparity changes of USPSTF 2021 guidelines and ACS 2023 recommendations by race and ethnicity among participants who formerly smoked only. Accuracy change arrows compare subgroup changes across guidelines (eg, White American: USPSTF-ACS vs Native Hawaiian: USPSTF-ACS). Disparity change arrows compare accuracy changes across subgroups (eg, USPSTF: Native Hawaiian-White American vs ACS: Native Hawaiian-White American). A, Accuracy change comparing Native Hawaiian participants who formerly smoked vs White American participants who formerly smoked. B, Disparity change comparing Native Hawaiian participants who formerly smoked vs White American participants who formerly smoked. C, Accuracy change comparing Black or African American participants who formerly smoked vs White American participants who formerly smoked. D, Disparity change comparing Black or African American participants who formerly smoked vs White American participants who formerly smoked. E) Accuracy change comparing Hispanic or Latino participants who formerly smoked vs White American participants who formerly smoked. F, Disparity change comparing Hispanic or Latino participants who formerly smoked vs White American participants who formerly smoked. G, Accuracy change comparing Japanese American participants who formerly smoked vs White American participants who formerly smoked. H, Disparity change comparing Japanese American participants who formerly smoked vs White American participants who formerly smoked. AA = African American; ACS = American Cancer Society; USPSTF = US Preventive Services Task Force.*

National Lung Screening Trial, and the Prostate, Lung, Colorectal, and Ovarian Cancer Screening trial).[11-13] Our study contextualizes these recommendations by assessing their effect on racial and ethnic disparities in a large racially and ethnically diverse study population. Overall, we found that the ACS recommendations increased screening eligibility regardless of lung cancer status, which potentially could increase unnecessary screening of people at low risk of lung cancer.

Improving the true-positive rate (ie, increasing the number of people at risk of lung cancer who are included in screening) is crucial, but lung cancer screening is not risk free. The inherent risk to benefit trade-off associated with an increased number of LDCT tests must be considered carefully. One study evaluating potential harms of lung screening found that more than one-third of patients completing screening LDCT imaging underwent downstream imaging or had findings that required further diagnostic workup, including invasive procedures. One-fifth of individuals who underwent such invasive procedures experienced major complications.[18] For participants who develop lung cancer, the proposed changes could mean increased access to potentially life-saving screening. For participants who do not develop lung cancer, which constitutes the vast majority of the population that smokes (approximately 80%),[19] the proposed changes increase the likelihood of unnecessary screening, further diagnostic workup, and downstream complications with no apparent benefit.

Our data suggest that the ACS recommendations operate at a different point than the USPSTF guidelines on the same hypothetical receiver operating characteristic curve. At the time screening is implemented, we do not know who among those eligible for screening will develop lung cancer. Yet, we can increase the probability of screening such individuals by screening in a population with high

Figure 1 – *Continued*



disease prevalence. Ideally, lung cancer screening guidelines will prioritize both the inclusion of individuals at high risk of lung cancer and the exclusion of individuals at low risk of lung cancer. The ACS recommendations manage this balance at a new, although not necessarily better, sensitivity-specificity trade-off. As a case in point, although the ACS recommendations identified more cases of lung cancer, the overwhelming majority of newly eligible participants (97%) did not develop lung cancer.

Importantly, the change in the sensitivity-specificity trade-off was not uniform across racial and ethnic subgroups and led to wider disparities in screening eligibility and poorer screening efficiency. The ACS recommendations performed most inefficiently for White American participants and least inefficiently for Hispanic or Latino participants by overselecting White American individuals without lung cancer for screening. That is, we would expect to find more White American individuals at low risk of lung cancer to be eligible for screening compared with their Native Hawaiian, Japanese American, Black or AA, or Hispanic or Latino counterparts as a result of using the ACS recommendations instead of the USPSTF guidelines. The question remains whether this new trade-off between the potential benefits (more lung cancer cases identified) and risks (larger racial disparities and more unnecessary screening) is the optimal choice for screening the broader population.

Achieving a lung cancer screening strategy that eliminates racial and ethnic disparities in eligibility while maintaining high sensitivity and high specificity may require a different approach than the current 1-size-fits-all criteria. In this instance, it is not possible to maximize accuracy within every subgroup of interest using the same criteria or rules for each subgroup. One strategy to improve equity could be to apply a combination of the ACS recommendations and

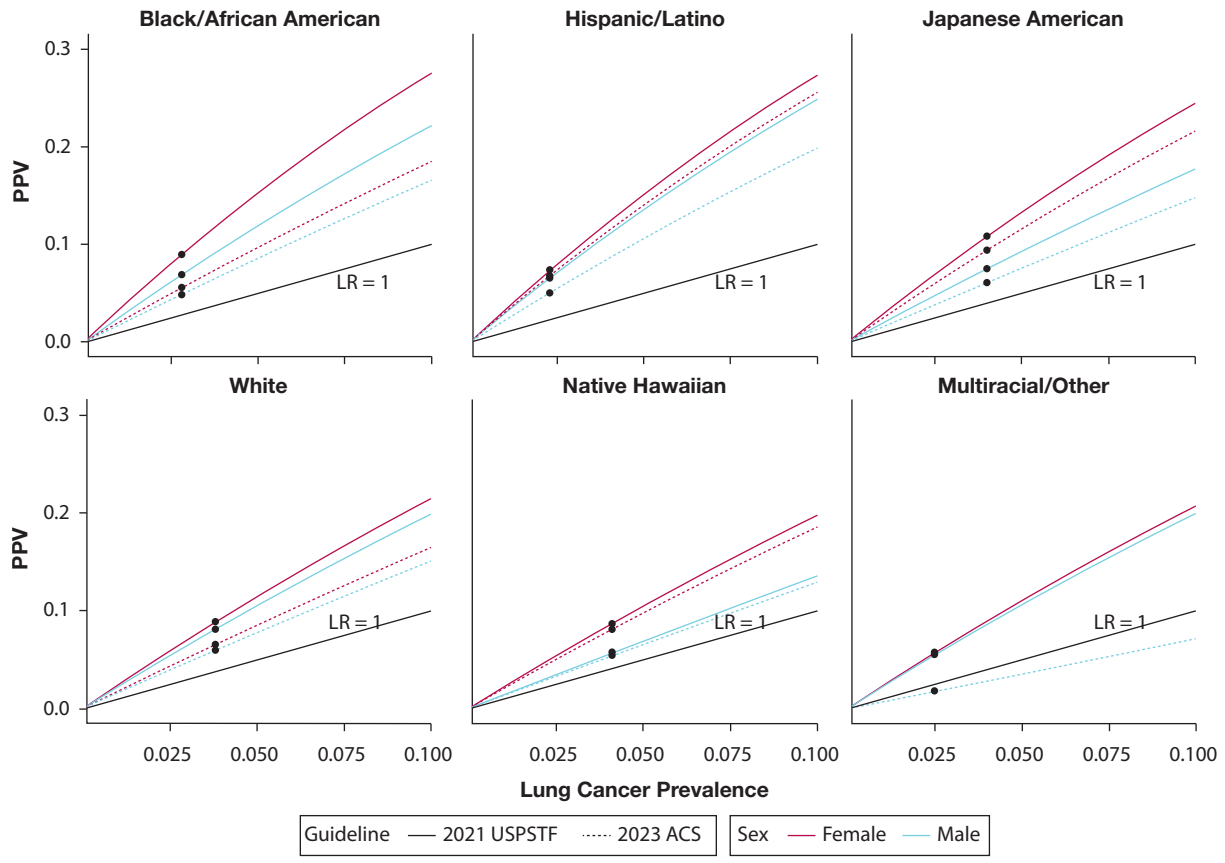


Figure 2 – *Positive LR curves showing the relationship between lung cancer prevalence and PPVs for USPSTF 2021 guidelines and ACS 2023 recommendations by race, ethnicity, and sex among participants who formerly smoked only. Positive LR measures the ratio of eligible people with lung cancer to eligible people without lung cancer. PPV measures the proportion of eligible participants who developed lung cancer. Points represent study-specific lung cancer prevalence. ACS = American Cancer Society; LR = likelihood ratio; PPV = positive predictive value; USPSTF = US Preventive Services Task Force.*

USPSTF guidelines. Yet, it is not a simple task to strike a meaningful balance between the 2 guidelines, because the end goals for population-level and group-level accuracy must be considered carefully and purposefully. We demonstrate this complexity with a simple example in e-Tables 3 and 4. In this illustrative combination, the USPSTF guidelines are applied to White American participants only and the ACS recommendations are applied to non-White American participants only. We show that this is a compromise between the 2 guidelines, but not necessarily optimal overall. Alternatively, expanding the current criteria to include additional lung cancer risk factors could improve screening efficiency. Risk model-based approaches that account for risk factors beyond age and smoking history have been shown to improve sensitivity and specificity and can reduce racial and ethnic eligibility inequities when applied in a manner that is tailored to subpopulations of interest.[20] A more extensive solution may involve incorporating other screening methods, such as blood-based biomarker tests, which would enrich the screening population with high-risk individuals whose characteristics are not captured by existing models or criteria. Ultimately, the measure of the appropriateness of our lung cancer screening eligibility strategy should be its ability to manage the precarious balance among high diagnostic accuracy, low ease of application, and minimal inequities across varying populations and geographic regions.

This large and racially diverse cohort of adults who had ever smoked with approximately 26 years of follow-up was ideal for this analysis. However, we must acknowledge some limitations. First, our smoking measures were self-reported and required estimating some pack-years using the last recorded smoking status and cigarettes smoked per day (last observation carried forward) to complete missing exposure data. Given the stability of smoking patterns over time, this is unlikely to be a major statistical concern. In practice, the same approach will be required because most health care systems

implementing these guidelines also rely on self-reported data. Also, our multiracial or other group combines participants from heterogeneous backgrounds with known differences in lung cancer risk. Thus, the benefits or harms shown for this group represent an average effect that may not apply to any 1 subgroup. Finally, our measure of quit duration was calculated assuming no recidivism. Given that the ACS recommendations are applicable only for individuals who formerly smoked, this may result in overestimation of sensitivity and specificity. However, research has shown that the odds of relapse after smoking cessation decrease with increasing years since quitting,[21] so large amounts of recidivism are unlikely.

Interpretation

The ACS recommendations led to increased screening eligibility regardless of lung cancer status, potentially increasing unnecessary screening of people without lung cancer. The magnitude of the change was not uniform across racial and ethnic subgroups. The ACS recommendations disproportionately increased eligibility among White American participants who formerly smoked compared with their Japanese American, Native Hawaiian, Black or AA, or Hispanic or Latino counterparts, resulting in larger racial, ethnic, and sex disparities in eligibility. Our analysis demonstrated that racial, ethnic, and sex disparities in screening eligibility remain and require redress.

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Additional information: The e-Figure and e-Tables are available online under "Supplementary Data."

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